项目 4 园区网络接入安全项目

【项目导学】

本项目构建了 DHCP 攻击及安全防范、AAA 的规划与配置、IMC 的安装与网络管理的规划与部署、802.1X 的规划与配置和 PORTAL 认证的规划与配置五个任务，学生通过学习，可以掌握使用网络安全防护、用户接入认证、网络设备集中管理等相关技术的规划与部署实施。

【学习目标】

通过本项目的学习，掌握智慧园区网络接入安全的规划与部署技术，实现以下课程目标达成。

技能目标

1.能独立完成 DHCP 攻击模拟与防护配置，通过抓包分析攻击流量，熟练部署 DHCP Snooping、DAI 等安全策略。

2.熟练运用 AAA 框架，完成 RADIUS 服务器对接与策略配置，实现用户接入的认证、授权、计费全流程实操。

3.掌握 IMC 软件安装部署，能通过 IMC 实现园区网络设备的批量配置、拓扑可视化管理及故障预警，完成网络集中管控。

4.精准规划 802.1X 认证架构，完成交换机、认证服务器、终端的联动配置，保障有线接入安全；独立搭建 PORTAL 认证环境，调试页面跳转、认证交互流程。

知识目标

1.理解 DHCP 协议工作原理，熟悉常见攻击（如仿冒服务器、饿死攻击）的原理与防范逻辑，掌握 DHCP 安全扩展技术（Snooping、Option82）的应用场景。

2.吃透 AAA 模型架构，明晰 RADIUS 协议交互流程，区分认证、授权、计费的功能边界，熟悉不同接入场景（有线、无线）的 AAA 部署方案。

3.掌握 IMC 网络管理平台的设计理念，理解拓扑发现、配置备份、性能监控等模块的实现机制，熟悉网络管理协议（SNMP、ICMP）的应用。

4.深度掌握 802.1X（EAP 协议体系）、PORTAL 认证的原理差异，熟悉认证触发条件、交互流程，了解不同认证方式的适用场景与优缺点。

素养目标

1.安全合规意识：严格遵循网络安全规范实施配置，重视用户数据、网络设备的安全防护，杜绝因操作失误引发安全漏洞，养成 “配置即合规” 的职业习惯。

2.问题解决能力：遇到网络故障（如认证失败、攻击防护失效），能通过日志分析、抓包工具、文档查阅等方式，精准定位问题根源并高效解决，培养 “拆解 - 分析 - 验证” 的思维闭环。

3.团队协作素养：在项目实施中（如 IMC 多模块部署、多认证方式协同），主动与团队成员分工配合，清晰传递技术方案、共享调试经验，提升团队整体交付效率。

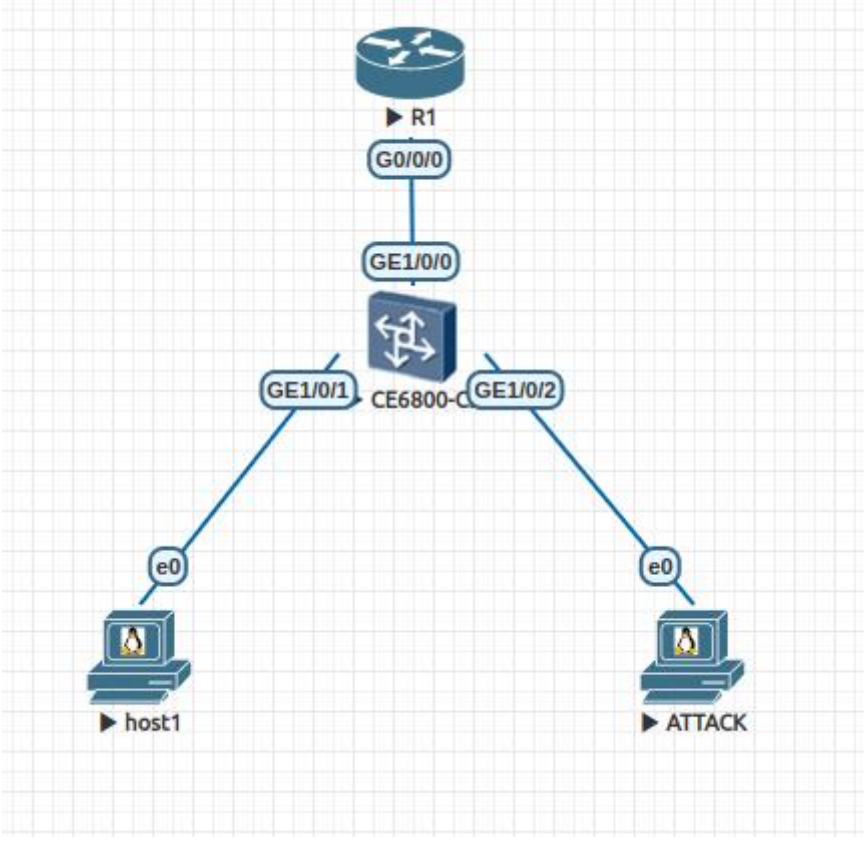
4.持续学习思维：紧跟园区网络安全技术迭代（如新型接入认证、AI 驱动的攻击防范），主动

探索新技术应用，保持对行业动态、技术创新的敏锐感知，构建终身学习的职业发展观。

【能力图谱】

任务 1 DHCP 攻击与安全防御

在网络环境中，DHCP（动态主机配置协议）负责为终端自动分配 IP 地址等网络参数，是网络通信的重要基础。然而，DHCP 协议存在一定的安全漏洞，攻击者可通过特定手段发起攻击，导致合法终端无法获取 IP 地址，进而影响网络正常运行。本任务围绕 DHCP 饿死攻击展开，通过模拟攻击与防御配置，帮助理解 DHCP 攻击原理及相应的安全防御措施。



（用户名：kali 密码：Key-1122）ATTACK 主机作为 dhcp 攻击发起者

➢ 任务实施

一. DHCP 饿死攻击及防御

1. 进入交换机 CE6800，选择配置方式。

```
提示：
配置的命令需要运行commit后生效。
不想要运行commit，可以在进入系统视图的时候使用system-view immediately进入。
system-view immediately可缩写为sys im

Info: The max number of VTY users is 5, the number of current VTY users online is 0, and to
tal number of terminal users online is 1.
      The current login time is 2025-07-23 08:09:15.
<HUAWEI>
<HUAWEI>
<HUAWEI>
<HUAWEI>
<HUAWEI>system-view immediately
Enter system view, return user view with return command.
[HUAWEI]
[HUAWEI]
```

2. 修改设备名称，开启交换机 DHCP 功能，配置 DHCP snooping 功能，防止非法的 DHCP 服务器分配 IP 地址。

```
[HUAWEI]sysname SW        # 修改设备名称
[SW]
[SW]
[SW]dhcp enable           # 开启DHCP功能
[SW]
[SW]
[SW]dhcp snooping  enable
[SW]                      # 启用DHCP snooping功能
[SW]
```

3. 将接入终端接口全部启用 DHCP Snooping 功能（端口过多时可以使用端口批量配置）

```
[SW]
[SW]interface  GE 1/0/1
[SW-GE1/0/1]        # 开启接口DHCP snooping功能
[SW-GE1/0/1]dhcp snooping enable
[SW-GE1/0/1]
[SW-GE1/0/1]quit
[SW]
[SW]interface  GE 1/0/2
[SW-GE1/0/2]
[SW-GE1/0/2]dhcp snooping enable
[SW-GE1/0/2]
```

4. 将连接 DHCP_Server 的端口设置为信任接口。

```
[SW]interface  GE 1/0/0
[SW-GE1/0/0]
[SW-GE1/0/0]
[SW-GE1/0/0]dhcp snooping trusted
[SW-GE1/0/0]
```

5. 登录路由器 R1，初始密码均为 super，第一次登录需要修改密码，重新修改密码为 gzqk@123

```
Username:super
Password:                # 账户与密码均为super
Warning: The password is already expired.
The password needs to be changed. Change now? [Y/N]: Y   # 重新修改密码
Please enter old password:
Please enter new password:
Please confirm new password:
The password has been changed successfully.
<Huawei>
Jul 23 2025 08:50:09+00:00 Huawei AAA/4/ChangePasswordAlarm:OID 1.3.6.1.4.1.2011.5.2.2
<Huawei>
Jul 23 2025 08:50:09+00:00 Huawei LINE/4/USERLOGIN:OID 1.3.6.1.4.1.2011.5.25.207.2.2 A
<Huawei>
<Huawei>
<Huawei>
```

6. 命名设备

```
<Huawei>sy
Enter system view, return user view with Ctrl+Z.
[Huawei]
[Huawei]sy
[Huawei]sysname RT
[RT]
[RT]
```

7. 配置路由器接口地址，作为终端的网关。

```
[RT]interface GigabitEthernet 0/0/0
[RT-GigabitEthernet0/0/0]
[RT-GigabitEthernet0/0/0]ip address 192.168.100.254 24
[RT-GigabitEthernet0/0/0]
```

8. DHCP 服务器配置，创建地址池，并分配地址给内网终端；启用接口采用全局地址池的 DHCP Server 功能。通过此命令，设备可以从全局 IP 地址池中为 DHCP 客户端分配合适的 IP 地址。

```
[RT]
[RT]ip pool ip-100          # 创建地址池 ip-100
[RT-ip-pool-ip-100]                          # 设置网关地址为192.168.100.254
[RT-ip-pool-ip-100]gateway-list 192.168.100.254
[RT-ip-pool-ip-100]
[RT-ip-pool-ip-100]network 192.168.100.0 mask 255.255.255.0
[RT-ip-pool-ip-100]                                  # 设置分配地址网段
[RT-ip-pool-ip-100]
[RT-ip-pool-ip-100]dns-list 114.114.114.114 8.8.8.8
[RT-ip-pool-ip-100]                          # 配置DNS服务器地址
[RT-ip-pool-ip-100]
[RT-ip-pool-ip-100]qu
[RT]
[RT]interface g0/0/0
[RT-GigabitEthernet0/0/0]   # 为当前接口指定DHCP模式为全局模式
[RT-GigabitEthernet0/0/0]dhcp select global
[RT-GigabitEthernet0/0/0]
```

9. 登录 2 台终端主机，连接网络，打开命令行终端，分别输入 ip addr，查看是否获取到 IP 地址。

```
┌──(kali㉿host-10-10-220-112)-[~/桌面]
└─$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 50:00:00:03:00:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.150/24 brd 192.168.100.255 scope global dynamic noprefix
route eth0
       valid_lft 86339sec preferred_lft 86339sec
    inet6 fe80::5200:ff:fe03:0/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kali㉿host-10-10-220-112)-[~/桌面]
```

```
kali@host-10-10-220-112: ~/桌面
文件  动作  编辑  查看  帮助
┌──(kali㉿host-10-10-220-112)-[~/桌面]
└─$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 50:00:00:04:00:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.10/24 brd 192.168.100.255 scope global dynamic noprefixr
oute eth0
       valid_lft 86350sec preferred_lft 86350sec
    inet6 fe80::5200:ff:fe04:0/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kali㉿host-10-10-220-112)-[~/桌面]
└─$
```

10.DHCP_Server 服务器查看 IP 地址分配情况。 .

```
<RT>
<RT>display  ip pool name  ip-100  used   # 查看DHCP分配的地址
  Pool-name         : ip-100
  Pool-No           : 0
  Lease             : 1 Days 0 Hours 0 Minutes
  Domain-name       : -
  DNS-server0       : 114.114.114.114
  DNS-server1       : 8.8.8.8
  NBNS-server0      : -
  Netbios-type      : -
  Position          : Local
  Status            : Unlocked
  Gateway-0         : 192.168.100.254
  Network           : 192.168.100.0
  Mask              : 255.255.255.0
  VPN instance      : --
  Logging           : Disable
  Conflicted address recycle interval: -
  Address Statistic: Total       :253       Used        :2
                     Idle        :251       Expired     :0
                     Conflict    :0         Disabled    :0

 ----------------------------------------------------------------------------
  Network section
         Start           End       Total    Used Idle(Expired) Conflict Disabled
 ----------------------------------------------------------------------------
   192.168.100.1 192.168.100.254     253       2        251(0)       0     0
 ----------------------------------------------------------------------------
 Client-ID format as follows:
   DHCP  : mac-address                 PPPoE   : mac-address
   IPSec : user-id/portnumber/vrf      PPP     : interface index
   L2TP  : cpu-slot/session-id         SSL-VPN : user-id/session-id
 ----------------------------------------------------------------------------
  Index              IP             Client-ID     Type       Left   Status
 ----------------------------------------------------------------------------
      9  192.168.100.10        5000-0004-0000    DHCP      86157   Used
    149 192.168.100.150        5000-0003-0000    DHCP      86064   Used
 ----------------------------------------------------------------------------
 ----------------------------------------------------------------------------
<RT>
```

11. 登录 ATTACK 终端主机，进入管理员模式。

```
┌──(kali㉿host-10-10-220-112)-[~/桌面]
└─$ su root
密码：
┌──(root㉿host-10-10-220-112)-[/home/kali/桌面]
└─#

┌──(root㉿host-10-10-220-112)-[/home/kali/桌面]
└─# cd

┌──(root㉿host-10-10-220-112)-[~]
└─#
```

进入管理员模式，密码Key-1122

12. 执行 dhcpstarv -v -i eth0 命令，启用 dhcpstarv 攻击，出接口为 eth0 接口(连接 DHCP 服务器的接口)。

dhcpstarv 攻击时，源 MAC 地址是不会改变的，只有 Chaddr 地址（报文内的描述地址）会变化。

```
└─# dhcpstarv -v -i eth0
Copyright (C) 2007 Dmitry Davletbaev
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it under
certain conditions; see <http://www.gnu.org/licenses/> for details.

19:11:05 07/23/25: setting eth0 to promiscuous mode
19:11:06 07/23/25: no network mask option in DHCPOFFER
19:11:06 07/23/25: got address 192.168.100.110 for 00:16:36:ce:a4:0e from 192
.168.100.254
19:11:07 07/23/25: no network mask option in DHCPOFFER
19:11:07 07/23/25: got address 192.168.100.35 for 00:16:36:23:1c:43 from 192.
168.100.254
19:11:08 07/23/25: no network mask option in DHCPOFFER
19:11:08 07/23/25: got address 192.168.100.145 for 00:16:36:e3:63:fb from 192
.168.100.254
19:11:09 07/23/25: no network mask option in DHCPOFFER
19:11:09 07/23/25: got address 192.168.100.33 for 00:16:36:fc:c1:69 from 192.
168.100.254
```

13. 这样修改 Chaddr 的情况，DHCP Snooping 并不能进行防御，我们可以看到 DCHP Snooping 绑定表也被占满了的情况。

```
[SW]display  user-bind dhcp snooping all
Flags: O - outer vlan, I - inner vlan, P - map vlan
IP Address       MAC Address     VSI/VLAN(O/I/P) Interface      Lease
--------------------------------------------------------------------------------
192.168.100.113  0016-3604-e362  1   /--  /--    GE1/0/2        2025-07-24 11:11:11
192.168.100.22   0016-361d-a001  1   /--  /--    GE1/0/2        2025-07-24 11:11:12
192.168.100.35   0016-3623-1c43  1   /--  /--    GE1/0/2        2025-07-24 11:11:07
192.168.100.18   0016-363c-9890  1   /--  /--    GE1/0/2        2025-07-24 11:11:10
192.168.100.125  0016-365a-ca19  1   /--  /--    GE1/0/2        2025-07-24 11:11:18
192.168.100.238  0016-365f-e9f9  1   /--  /--    GE1/0/2        2025-07-24 11:11:17
192.168.100.179  0016-3694-8678  1   /--  /--    GE1/0/2        2025-07-24 11:11:13
192.168.100.100  0016-36b5-29a9  1   /--  /--    GE1/0/2        2025-07-24 11:11:14
192.168.100.110  0016-36ce-a40e  1   /--  /--    GE1/0/2        2025-07-24 11:11:06
192.168.100.212  0016-36cf-dadc  1   /--  /--    GE1/0/2        2025-07-24 11:11:15
192.168.100.215  0016-36dc-c121  1   /--  /--    GE1/0/2        2025-07-24 11:11:16
192.168.100.145  0016-36e3-63fb  1   /--  /--    GE1/0/2        2025-07-24 11:11:08
192.168.100.66   0016-36f1-c1e5  1   /--  /--    GE1/0/2        2025-07-24 11:11:19
192.168.100.33   0016-36fc-c169  1   /--  /--    GE1/0/2        2025-07-24 11:11:09
192.168.100.150  5000-0003-0000  1   /--  /--    GE1/0/1        2025-07-24 11:02:09
192.168.100.10   5000-0004-0000  1   /--  /--    GE1/0/2        2025-07-24 11:03:42
--------------------------------------------------------------------------------
Print count:           16         Total count:           16
[SW]a
```

14.作为 DHCP Server 服务器也在疯狂的回复 DHCP Offer 报文，并分配 IP 地址,HCP_Server 服务器的 IP 地址也迅速被分配出去（测试的时候，不要攻击太久 会造成卡顿）

```
<RT>display  ip pool name  ip-100  used
  Pool-name        : ip-100
  Pool-No          : 0
  Lease            : 1 Days 0 Hours 0 Minutes
  Domain-name      : -
  DNS-server0      : 114.114.114.114
  DNS-server1      : 8.8.8.8
  NBNS-server0     : -
  Netbios-type     : -
  Position         : Local
  Status           : Unlocked
  Gateway-0        : 192.168.100.254
  Network          : 192.168.100.0
  Mask             : 255.255.255.0
  VPN instance     : --
  Logging          : Disable
  Conflicted address recycle interval: -
  Address Statistic: Total       :253       Used        :26
                     Idle        :227       Expired     :0
                     Conflict    :0         Disabled    :0

 ---------------------------------------------------------------------------
  Network section
         Start           End       Total    Used Idle(Expired) Conflict Disabled
 ---------------------------------------------------------------------------
   192.168.100.1 192.168.100.254     253      26        227(0)       0     0
 ---------------------------------------------------------------------------
 Client-ID format as follows:
   DHCP  : mac-address                 PPPoE   : mac-address
   IPSec : user-id/portnumber/vrf      PPP     : interface index
   L2TP  : cpu-slot/session-id         SSL-VPN : user-id/session-id
 ---------------------------------------------------------------------------
  Index              IP             Client-ID    Type       Left   Status
 ---------------------------------------------------------------------------
      9  192.168.100.10        5000-0004-0000    DHCP      85935   Used
     13  192.168.100.14        0016-3662-cf66    DHCP      86400   Used
     17  192.168.100.18        0016-363c-9890    DHCP      86382   Used
     21  192.168.100.22        0016-361d-a001    DHCP      86384   Used
     22  192.168.100.23        0016-36ce-ffb3    DHCP      86394   Used
     32  192.168.100.33        0016-36fc-c169    DHCP      86381   Used
     34  192.168.100.35        0016-3623-1c43    DHCP      86379   Used
     37  192.168.100.38        0016-36a3-4c50    DHCP      86399   Used
     65  192.168.100.66        0016-36f1-c1e5    DHCP      86391   Used
     77  192.168.100.78        0016-36ba-2aab    DHCP      86395   Used
     99 192.168.100.100        0016-36b5-29a9    DHCP      86386   Used
    109 192.168.100.110        0016-36ce-a40e    DHCP      86378   Used
    112 192.168.100.113        0016-3604-e362    DHCP      86383   Used
    116 192.168.100.117        0016-3693-4af0    DHCP      86398   Used
```

15. hcpstarv 攻击时，源 MAC 地址是不变的，只有 Chaddr 地址是在变化的，所以我们在 SW 上开启 Snooping Check 检查，检查源 MAC 与 Chaddr 是否一致（一致则通过，不一致则丢弃）

```
[SW]interface  GE 1/0/1
[SW-GE1/0/1]
[SW-GE1/0/1]dhcp snooping check mac-address  enable
[SW-GE1/0/1]
[SW-GE1/0/1]
[SW-GE1/0/1]interface  GE 1/0/2
[SW-GE1/0/2]
[SW-GE1/0/2]
[SW-GE1/0/2]dhcp snooping check mac-address  enable
[SW-GE1/0/2]
[SW-GE1/0/2]
```

16. 我们先清空 DHCP 地址池

<RT> reset ip pool name ip-100 all

```
<RT>
<RT>reset iP pool name ip-100 all
Warning: If the IP addresses that are being used are reclaimed, may influence normal user in the network. Are you sure to continue?[Y/N]:y
<RT>
<RT>
<RT>
```

```
[RT]display  ip pool name ip-100
  Pool-name         : ip-100
  Pool-No           : 0
  Lease             : 1 Days 0 Hours 0 Minutes
  Domain-name       : -
  DNS-server0       : 114.114.114.114
  DNS-server1       : 8.8.8.8
  NBNS-server0      : -
  Netbios-type      : -
  Position          : Local
  Status            : Unlocked
  Gateway-0         : 192.168.100.254
  Network           : 192.168.100.0
  Mask              : 255.255.255.0
  VPN instance      : --
  Logging           : Disable
  Conflicted address recycle interval: -
  Address Statistic: Total       :253       Used        :0
                     Idle        :253       Expired     :0
                     Conflict    :0         Disabled    :0

 ----------------------------------------------------------------------------
  Network section
         Start           End       Total    Used Idle(Expired) Conflict Disabled
 ----------------------------------------------------------------------------
   192.168.100.1 192.168.100.254     253       0        253(0)       0     0
 ----------------------------------------------------------------------------
```

17. ATTACK 再进行 dhcpstarv 攻击，攻击已经失效，DHCP 服务器并不会继续分配 IP 地址。

```
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This is free software, and you are welcome to redistribute it under
certain conditions; see <http://www.gnu.org/licenses/> for details.

19:53:55 07/23/25: timeout while waiting for incoming data (2 seconds)
19:53:57 07/23/25: timeout while waiting for incoming data (2 seconds)
19:53:59 07/23/25: timeout while waiting for incoming data (2 seconds)
19:53:59 07/23/25: did not received DHCP reply for lease request (DHCPDISCOVE
R) in 3 retries
19:54:01 07/23/25: timeout while waiting for incoming data (2 seconds)
19:54:03 07/23/25: timeout while waiting for incoming data (2 seconds)
19:54:05 07/23/25: timeout while waiting for incoming data (2 seconds)
19:54:05 07/23/25: did not received DHCP reply for lease request (DHCPDISCOVE
R) in 3 retries
19:54:07 07/23/25: timeout while waiting for incoming data (2 seconds)
19:54:09 07/23/25: timeout while waiting for incoming data (2 seconds)
^C19:54:09 07/23/25: Exit.

┌──(root㉿host-10-10-220-112)-[~]
└─#
```

18. 重新查看地址分配情况，已经有效防御，不会分配随便分配地址。

```
[RT]display  ip pool name ip-100
  Pool-name         : ip-100
  Pool-No           : 0
  Lease             : 1 Days 0 Hours 0 Minutes
  Domain-name       : -
  DNS-server0       : 114.114.114.114
  DNS-server1       : 8.8.8.8
  NBNS-server0      : -
  Netbios-type      : -
  Position          : Local
  Status            : Unlocked
  Gateway-0         : 192.168.100.254
  Network           : 192.168.100.0
  Mask              : 255.255.255.0
  VPN instance      : --
  Logging           : Disable
  Conflicted address recycle interval: -
  Address Statistic: Total       :253       Used        :0
                     Idle        :253       Expired     :0
                     Conflict    :0         Disabled    :0

 ----------------------------------------------------------------------------
  Network section
         Start           End       Total    Used Idle(Expired) Conflict Disabled
 ----------------------------------------------------------------------------
   192.168.100.1 192.168.100.254     253       0        253(0)       0     0
 ----------------------------------------------------------------------------
[RT]
```

二.DHCP 饿死攻击及防御(扩展)

攻击者仅修改 Chaddr 进行 DHCP 攻击时,我们可以使用 Chaddr Check 检查,检查 S-MAC 与 Chaddr 地址是否相同，相同则通过，不相同则丢弃。若是 Chaddr 和 S-MAC 同时修改时，Chaddr Check 检查并不能起到作用，那我们该怎么办呢？

1.我们可以在 SW 开启端口安全，设置最大 MAC 地址数为 1，并配置超出关闭端口。

```
[SW]
[SW]interface GE 1/0/1
[SW-GE1/0/1]
[SW-GE1/0/1]port-security  enable        #开启端口安全功能
[SW-GE1/0/1]
[SW-GE1/0/1]port-security maximum 1      #设置端口最大mac地址数量
[SW-GE1/0/1]
[SW-GE1/0/1]port-security  protect-action error-down
Info: If the interface is triggered to error-down state, you need
[SW-GE1/0/1]                             #配置超出数量则关闭端口
[SW-GE1/0/1]quit
[SW]
[SW]mac-address  aging-time 60
[SW]
[SW]
[SW]
[SW]
```